MIL-P-63139A (AR)
1 May 1979
SUPERSEDING
MIL-P-63139(AR)
1 August 1977

MILITARY SPECIFICATION
PROJECTILE, 8-INCH, HE, RA, M650
DELAY ASSEMBLY, PARTS FOR

This Specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Delay Assembly Parts for use in Projectile, 8-Inch, HE, RA, M650.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this Military Specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 -Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

DRAWINGS

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9287957 - Plug, Igniter
9287958 - Washer, Delay
9287972 - Housing, Delay

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with Dwg. 9287957, 9287958, 9287972 and shall comply with all requirements of applicable specifications.

3.2 Workmanship.- The requirements for workmanship are as shown on the applicable drawing, in the applicable specification and the following:

3.2.1 Surface Preparation.- Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.2.2 Finishing.- Finishing shall comply with the requirements of the applicable drawing. All finishes shall be dry to the touch before packing for shipment.

3.2.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.2.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.2.5 Foreigh Matter.- No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.3 First Article Inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

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4.2 Classification of Inspections.- The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.- The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Housing, Delay	9287972	10
Plug, Igniter	9287957	10 (Consecutively produced from each cavity of each mold.)
Washer, Delay	9287958	10

4.3.2 Inspections to be Performed.- See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.- See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Delay Assembly Parts For				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Housing, Delay (DWG. 9287972) Examination for Defects	10	--	3.3	4.4.2.1
	Washer Delay (DWG. 9287958) Examination for Defects	10	--	3.3	4.4.2.2
	Plug, Igniter (DWG. 9287957) Examination for Defects	10	--	3.3	4.4.2.3

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.- See MIL-A-48078.

a. Sampling Plans.- Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
4.4.2.1	Housing Delay				9287972
					NEXT HIGHER ASSEMBLY 9287970
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Two Delay Column Holes do not break through counterbore at tube		100%	3.1	Visual
102	Minor thread diameter		0.40%	3.1	Gage
103	True position of thread minor diameter to thread counterbore		0.40%	3.1	Gage
104	Diameter of thread counterbore		0.40%	3.1	Gage
105	Surface finish of thread counterbore		0.40%	3.1	Visual
106	Minimum effective thread		0.40%	3.1	Gage
107	Diameter of two column holes		0.40%	3.1	Gage
108	Knurl missing		0.40%	3.1	Visual
109	Location of small basic diameter on taper		0.40%	3.1	Gage
110	Location of large basic diameter on taper		0.40%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9287972
					NEXT HIGHER ASSEMBLY
					9287970
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Circular runout of small basic diameter		0.40%	3.1	Gage
112	Circular runout of large basic diameter		0.40%	3.1	Gage
113	Depth of O-ring groove		0.40%	3.1	Gage
114	Diameter of counterbore hole intersecting two column holes		0.40%	3.1	Gage
115	Outside diameter of tube		0.40%	3.1	Gage
116	Protective finish on outside surface		0.40%	3.1	Visual
117	Length of two column holes		0.40%	3.1	Gage
118	No sharp corners on column holes (2 places)		0.40%	3.1	Visual
119	True position of surface prior to knurling		0.40%	3.1	Gage
120	Depth of aft cavity		0.40%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A(AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9287972
					NEXT HIGHER ASSEMBLY
					9287970
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	True position of thread counterbore with tube outside diameter		0.65%	3.1	Gage
202	Chamfer at thread counterbore missing		0.65%	3.1	Visual
203	True position of 2 column holes to tube		0.65%	3.1	Gage
204	Outside diameter of step below knurled surface		0.65%	3.1	Gage
205	Diameter of surface before knurling		0.65%	3.1	Gage
206	Distance large end surface to chamfer at large outside diameter		0.65%	3.1	Gage
207	Width of O-ring groove		0.65%	3.1	Gage
208	Location of O-ring groove		0.65%	3.1	Visual
209	Corner radii of O-ring groove missing		0.65%	3.1	Visual
210	Surface finish of O-ring groove		0.65%	3.1	Visual
211	Overall length		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSEPCTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9287972
					NEXT HIGHER ASSEMBLY 9287970
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
212	Distance large end surface to shoulder at tube		0.65%	3.1	Gage
213	Distance end of tube to web between two column holes		0.65%	3.1	Gage
214	Distance end of tube to chamfer at bottom of tube		0.65%	3.1	Gage
215	True position of counterbore intersecting 2 column holes		0.65%	3.1	Gage
216	Wall thickness of tube, min.		0.65%	3.1	Gage
217	Wall thickness of tube at groove		0.65%	3.1	Gage
218	Angle of groove in tube		0.65%	3.1	Gage
219	Location of groove in tube		0.65%	3.1	Gage
220	Chamfer missing on tube, aft end, or on knurl surface		0.65%	3.1	Visual
221	Depth of thread counterbore		0.65%	3.1	Gage
222	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Washer, Delay		9287958
			NEXT HIGHER ASSEMBLY 9287970

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter		0.40%	3.1	Gage
102	Inside diameter		0.40%	3.1	Gage
103	True position of inside diameter		0.40%	3.1	Gage
104	Thickness		0.40%	3.1	Gage
Minor					
201	Protective finish missing or incomplete		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63139A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plug, Igniter				9287957
					NEXT HIGHER ASSEMBLY
					9287970
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outer diameter		(a)	3.1	Gage
102	Inner diameter		(a)	3.1	Gage
103	Angle of outside surface		(a)	3.1	Gage
104	Thickness of base		(a)	3.1	Gage
Minor					
201	Length		(a)	3.1	Gage
202	Angle of inside surface		(a)	3.1	Gage
203	Evidnece of poor workmanship		(a)	3.2	Visual

NOTES:

(a) See 4.4.3.1

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4.3.3 Testing

4.4.3.1 Molded Components.- The first 20 consecutively produced pieces from each cavity of each mold shall be inspected for all defects listed in paragraph 4.4.2.3. Additionally, after production of 2000 pieces per cavity per mold, 10 additional pieces from each cavity of each mold shall be consecutively produced and inspected for all defects listed in paragraph 4.4.2.3. If any major defect is found the production represented by the sample shall be rejected. Production shall be stopped and the mold corrected. If any minor defect is found, corrective action will be taken. During production, the molding procedures shall be monitored for uniformity of molding practices. Suspected deviation in mold product shall be examined to determine if corrective action or mold repair is required. If during production, correction or repair is required, 20 consecutively produced units from each affected mold cavity shall be inspected and shall comply with all drawing and specification requirements.

4.4.4 Inspection Equipment.- The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.- Packaging, packing and marking shall be in accordance with Level B of MIL-STD-1169.

6. NOTES

6.1 Intended Use.- The components covered by this specification are intended for use on the 8-Inch, M650, RA Projectile.

6.2 Ordering Data.- See MIL-A-48078(PA).

6.3 Submission of Inspection Procedures and/or Equipment Desgins for Approval.- See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

Custodian:
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Preparing Activity:
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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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